

ZHUMADIAN, China

WMO: 572900

Lat: 32.9403N

Lon: 113.9144E

Elev: 352

StdP: 14.51

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.0	25.6	1.4	6.0	33.5	6.7	7.9	34.8	16.6	42.5	14.3	41.7	2.9	270	0.370

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	95.8	78.2	93.6	77.3	91.4	76.7	82.8	91.3	81.5	89.6	80.3	87.8	6.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.7	161.7	88.0	79.3	154.2	86.8	78.2	148.2	85.5	47.3	91.3	45.8	89.8	44.4	87.8	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
14.0	11.9	10.3	DB	17.0	99.8	4.1	2.3	14.0	101.4	11.6	102.8	9.3	104.1	6.3	105.8
			WB	15.0	84.8	3.8	1.4	12.3	85.8	10.1	86.6	8.0	87.4	5.3	88.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	60.8	35.7	41.0	51.1	61.7	71.3	79.0	82.0	80.0	72.2	63.3	50.9	40.0	(6)
(7)		DBStd	17.15	5.93	7.11	7.91	7.55	6.19	5.28	4.88	5.27	5.57	6.44	7.48	5.99	(7)
(8)		HDD50	1185	444	262	81	7	0	0	0	0	1	76	314		(8)
(9)		HDD65	3500	908	673	435	148	16	1	0	9	109	426	775		(9)
(10)		CDD50	5122	1	9	116	359	661	871	992	929	665	414	101	4	(10)
(11)		CDD65	1959	0	0	4	50	211	422	527	464	223	57	1	0	(11)
(12)		CDH74	19544	0	0	35	440	1874	4514	6007	4706	1616	339	13	0	(12)
(13)		CDH80	8089	0	0	5	110	684	2033	2728	1928	531	69	1	0	(13)
(14)	Wind	WSAvg	4.6	4.6	5.1	5.6	5.2	4.9	4.4	3.9	3.8	3.9	4.5	4.6		(14)
(15)	Precipitation	PrecAvg	32.7	0.7	1.0	1.6	2.2	3.3	4.2	7.2	5.2	3.3	2.0	1.5	0.6	(15)
(16)		PrecMax	44.9	2.3	2.6	3.2	4.1	8.4	8.6	13.4	10.1	7.4	5.2	3.8	2.1	(16)
(17)		PrecMin	22.7	0.1	0.1	0.2	0.5	0.6	1.8	2.6	1.5	0.4	0.2	0.0	0.0	(17)
(18)		PrecStd	5.2	0.6	0.6	0.9	0.9	1.6	1.5	2.7	2.0	1.9	1.5	1.1	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.5	66.3	77.9	88.1	94.4	98.5	98.1	98.3	93.9	86.2	75.9	62.1	(19)
(20)		0.4%	MCWB	45.6	53.5	62.1	69.1	68.7	73.1	81.3	81.9	75.8	65.8	58.9	47.9	(20)
(21)		2%	DB	53.1	60.7	73.0	82.8	90.3	95.1	95.8	93.9	89.2	81.3	71.0	56.8	(21)
(22)		2%	MCWB	42.4	48.5	58.7	66.2	68.3	73.4	81.0	80.7	73.5	63.3	56.2	45.1	(22)
(23)		5%	DB	49.2	56.6	68.5	79.1	87.1	92.6	93.5	91.3	85.6	77.7	66.5	52.9	(23)
(24)		5%	MCWB	40.2	46.1	56.0	64.4	68.1	73.2	80.6	79.7	71.4	62.0	54.3	43.0	(24)
(25)		10%	DB	45.6	52.9	64.2	75.0	83.9	90.0	91.2	88.8	82.6	74.0	62.3	49.3	(25)
(26)		10%	MCWB	38.2	44.3	53.3	61.9	67.3	72.4	79.6	78.3	69.7	61.0	52.4	41.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.3	55.0	64.9	72.4	75.2	80.9	84.6	84.5	80.5	70.1	61.6	49.8	(27)
(28)		0.4%	MCDWB	55.3	63.4	74.5	82.3	84.8	90.9	93.2	92.5	90.6	76.4	71.4	59.1	(28)
(29)		2%	WB	43.6	50.5	60.6	68.8	73.1	79.0	83.2	82.8	77.4	67.0	58.4	47.2	(29)
(30)		2%	MCDWB	51.4	58.1	70.5	79.2	82.8	88.5	92.0	90.8	84.2	75.3	67.5	53.7	(30)
(31)		5%	WB	41.2	47.7	57.2	66.1	71.2	77.4	82.0	81.4	74.7	65.2	55.9	44.7	(31)
(32)		5%	MCDWB	47.9	54.0	66.4	75.6	81.4	86.5	90.5	88.7	80.9	73.2	63.6	50.8	(32)
(33)		10%	WB	38.9	45.5	54.4	63.7	69.4	75.8	80.8	80.1	72.4	63.4	53.4	42.6	(33)
(34)		10%	MCDWB	44.4	51.8	62.8	73.0	79.6	84.7	89.0	86.5	78.7	70.8	60.5	48.1	(34)
(35)	Mean Daily Temperature Range		MDBR	13.2	14.3	15.7	17.3	17.4	16.3	12.9	12.8	14.9	14.9	14.2	13.0	(35)
(36)		5% DB	MCDBR	20.4	21.1	22.1	23.4	23.1	20.7	16.3	16.1	20.2	21.2	21.5	20.5	(36)
(37)		5% DB	MCWBR	12.4	12.3	12.3	12.1	8.6	6.6	5.8	6.0	7.1	8.4	10.5	11.8	(37)
(38)		5% WB	MCDBR	18.8	18.2	20.7	20.6	18.6	15.6	14.4	14.0	15.7	16.9	18.3	17.4	(38)
(39)		5% WB	MCWBR	12.1	11.5	12.6	12.2	9.0	6.4	6.4	6.1	7.3	7.9	10.1	10.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.623	0.681	0.685	0.678	0.685	0.760	0.724	0.750	0.717	0.711	0.667	0.612	(40)	
(41)		taud	1.566	1.498	1.512	1.559	1.593	1.494	1.596	1.534	1.560	1.514	1.548	1.603	(41)	
(42)		Ebn at Noon	175	183	199	210	211	195	201	192	189	173	163	168	(42)	
(43)		Edh at Noon	71	83	88	87	85	94	84	88	83	81	71	65	(43)	
(44)	All-Sky Solar Radiation	RadAvg	688	813	1193	1422	1512	1517	1438	1343	1126	932	757	659	(44)	
(45)		RadStd	96	167	116	132	123	136	183	193	167	126	115	111	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+121	N/A	(47)

Nomenclature: See separate page